12-Week Junior Data Analyst Roadmap

Prepared 14 October 2026

A plan for a complete beginner, about 6 hours a week (roughly 72 hours), from spreadsheets to a public portfolio project. Every resource carries a cost tag.

How to read this plan

Each resource was opened before it was listed. No course names, links, certificates or prices were invented.

Tag	Meaning
Free	The page states it is free or no-cost, or it is open-source documentation.
Free (verify)	The page loaded but its content or pricing was not confirmed. Check it before you start.
Paid / trial	Costs money or is free only for a limited time. Optional.

The time budget is about 6 hours a week (the 5-7 hour range). The weekly Time values are estimates, not figures from the providers.

Phase 1: Spreadsheets and SQL (Weeks 1-4)

Week	Topics and objective	Resources	Tools	Exercise	Time	Completion criterion
1	Analyst workflow (ask, clean, analyse, share), spreadsheet basics, formatting, basic formulas. Objective: build and format a clean table.	Free: How to use Google Sheets, Edit and format, Formulas and functions. Free (verify): needs a Google account.	Google Sheets	Build a personal-expense or study-hours tracker with at least 30 rows. Add SUM, AVERAGE, COUNT and IF formulas.	6 h	At least 30 rows and at least 4 working formulas, with no error cells.
2	Data cleaning in a sheet, sorting and filtering, lookups, pivot tables, charts. Objective: turn a messy table into a summary.	Free: Google Sheets Help (same hub). Search it for pivot table and VLOOKUP; those specific article links were not verified.	Google Sheets	Download a small CSV from Kaggle Datasets (free to browse and download; check each dataset's licence). Clean it, build a pivot table and make 2 charts.	6 h	1 pivot table and 2 charts, and 3 written questions answered from the data.
3	SQL basics: SELECT, WHERE, ORDER BY, LIMIT, NULLs, expressions. Objective: write single-table queries.	Free (verify): SQLBolt, Lessons 1-9. The page shows no pricing, so free is assumed.	Browser (SQLBolt)	Complete SQLBolt Lessons 1-9. Then write 5 of your own questions about your Week 2 dataset in plain English.	6 h	All exercises in Lessons 1-9 marked complete.
4	Aggregates (COUNT, SUM, GROUP BY, HAVING), JOINs, subqueries. Objective: answer multi-table business questions.	Free (verify): SQLBolt aggregates, JOINs, subqueries. Free: Kaggle: Intro to SQL (uses Google BigQuery).	SQLBolt, Kaggle	Finish SQLBolt Lessons 10-12 and the JOIN and subquery topics. Start Kaggle Intro to SQL.	7 h	You can write a query using JOIN + GROUP BY + HAVING without a reference.

Phase 2: Python and Pandas (Weeks 5-7)

Week	Topics and objective	Resources	Tools	Exercise	Time	Completion criterion
5	Variables, lists, dictionaries, loops, functions. Objective: read and write basic Python.	Free: Kaggle: Python	Kaggle Notebooks	Complete the Kaggle Python exercises. Then write a function that takes a list of numbers and returns the min, max and mean.	6 h	All Kaggle Python exercises submitted, and your function runs correctly on 3 test inputs.
6	DataFrames, read_csv, selecting, filtering, groupby, merging. Objective: replicate your SQL and spreadsheet answers in Pandas.	Free: Kaggle: Pandas, 10 minutes to pandas (official docs).	Kaggle Notebooks	Load your Week 2 dataset and re-answer your 3 spreadsheet questions with Pandas.	6 h	The Pandas answers match your spreadsheet and SQL answers.
7	Missing values, duplicates, data types, text cleaning. Objective: clean a real, messy dataset.	Free: Kaggle: Data Cleaning, Pandas docs on missing data.	Kaggle Notebooks	Pick a new messy dataset. Document each cleaning step (what was wrong, what you did, and why) in markdown cells.	6 h	The notebook runs top to bottom, has at least 5 documented cleaning steps and leaves zero unexplained nulls.

Phase 3: Statistics and visualisation (Weeks 8-9)

Week	Topics and objective	Resources	Tools	Exercise	Time	Completion criterion
8	Mean, median, standard deviation, distributions, sampling, correlation, intro to confidence intervals. Objective: interpret summary statistics correctly.	Free (verify): Khan Academy: Statistics and probability; the page did not render, so confirm the units yourself. Free: describe() is shown in 10 minutes to pandas.	Kaggle Notebooks	Run describe() and a correlation check on your cleaned dataset. Write 5 sentences on what the numbers mean, including one limitation.	6 h	You can explain mean vs median, standard deviation and correlation vs causation in your own words, with examples from your data.
9	Choosing chart types (line, bar, histogram, scatter, heatmap), clear labelling, avoiding misleading charts. Objective: produce readable, honest charts.	Free: Kaggle: Data Visualization. Free (verify): optional Tableau Public; the page needs JavaScript, so its cost was not confirmed.	Kaggle Notebooks	Create 4 different chart types from your dataset. Give each a title, axis labels and a one-sentence takeaway.	6 h	4 charts, each with a title, labelled axes and a takeaway, and you can say why you chose each chart type.

Phase 4: Portfolio project (Weeks 10-12)

Week	Topics and objective	Resources	Tools	Exercise	Time	Completion criterion
10	Defining a business question, choosing a dataset, cleaning. Objective: set up the project.	Free: Kaggle Datasets; check the licence of the dataset you pick.	Kaggle Notebooks, Python, Pandas	Write a 1-paragraph problem statement with 3-4 specific questions. Load and clean the dataset.	6 h	The problem statement is written and the cleaned data is saved.
11	Analysis, SQL or Pandas queries, statistics, visuals, findings. Objective: answer your questions with evidence.	Reuse the Weeks 3-9 resources above.	Notebook, Sheets (optional)	Answer each question with at least one chart or table. Write 3-5 insights and 1-2 recommendations.	7 h	Every question from Week 10 has an answer backed by data.
12	Documentation, storytelling, publishing. Objective: present the work so others can read it.	Free (verify): GitHub free account; current plan details were not verified.	GitHub, Notebook	Write a README (problem, data source, method, findings, limitations). Publish the notebook. Ask someone to review it.	6 h	The repo is public. A reader can follow the README without help, and the notebook re-runs without errors.

Paid courses, trials and certificates

The Google Data Analytics Professional Certificate does not fit a 12-week, 6-hour-a-week plan, so treat it as optional.

Item	What was verified	Fits this plan?
Paid / trial: Google Data Analytics Professional Certificate (Coursera)	9 courses covering spreadsheets, SQL, Tableau and Python; over 180 hours of instruction, listed as 6 months at 10 hours a week. The FAQ says US/Canada pricing is $49 per month after a 7-day free trial, with a lower cost in some other countries. The page also shows Included with Coursera Plus and Enroll for free, but what that free enrolment includes was not confirmed.	No. It is longer than this plan. Verify your local price and what Enroll for free includes.
Paid / trial: Coursera Plus	The page mentions a subscription and a trial. Terms and price were not verified.	Optional.
Paid certificates in general	Not needed for this plan.	A public portfolio matters more than a certificate.

Skills to demonstrate after 12 weeks

- **Spreadsheets:** clean a table, use formulas and lookups, build pivot tables and simple charts.
- **SQL:** write queries with filters, aggregates, JOINs and subqueries.
- **Python:** write basic functions and loops, and read other people's simple scripts.
- **Pandas:** load, filter, group, merge and clean data, including missing values and duplicates.
- **Statistics:** explain and compute mean, median, standard deviation and correlation, and say why correlation is not causation.
- **Visualisation:** pick an appropriate chart, label it properly, and avoid misleading scales.
- **Communication:** turn analysis into 3-5 plain-language insights with a recommendation.
- **Portfolio:** one public GitHub project with a README, a documented notebook and a stated problem.

Items to verify yourself

- Khan Academy statistics content and Tableau Public cost (the pages did not render).
- That SQLBolt and GitHub are free for your use (no pricing was shown on the pages opened).
- Kaggle course durations (not shown on the pages) and each Kaggle dataset's licence.
- Whether you need a Google account for Sheets, and what limits it has.
- Your local Coursera price and what Enroll for free includes, if you consider the paid route.

Sources

Pages opened on 14 Oct 2026; some rendered only partially (see the verification list).

- Kaggle Learn

- SQLBolt
- Google Sheets Help
- Coursera: Google Data Analytics
- pandas documentation: 10 minutes to pandas
- Kaggle Datasets